

TIEFA Project Phase 1 – VLA Data Collection Protocol v1.0 & Task Realignment

Team,

As discussed in today's meeting, while we are waiting for the arrival of our physical robotic gripper, we will proceed with a "Software-First" strategy.

We will use this period to calibrate our pipeline and master the ****LoRA (Low-Rank Adaptation)**** fine-tuning process.

To ensure our 4B Vision-Language model trains successfully on the RTX machine without memory or alignment errors, all modules must strictly adhere to the following data collection protocol.

****Manual renaming or stitching of data is strictly prohibited.****

📁 Dataset Directory Structure

Module B (Vision & AI) must receive raw data organized into independent folders, where each folder represents a single grasp attempt (Episode). The directory must strictly follow this format:

```
TIEFA_Phase1_Dataset/
├── task_20260605_113000/      # Folder name format: task_YYYYMMDD_HHMMSS
(Auto-generated)
|   ├── image.jpg             # Pure, unedited raw RGB image from D435i
|   ├── instruction.txt        # Speech-to-text text instruction for this specific task
|   └── pose_action.json       # 6D posture data of the LARA arm's flange
├── task_20260605_113512/
|   ├── image.jpg
|   └── ...
...
...
```

🤝 Multi-Module Cross-Functional Pipeline

Every single data collection cycle must be executed in this exact sequential order:

1. Module C (HRI & Gateway) - The Initiator & Commander

*****Action:**** The student gives a verbal command (e.g., ****"Move LARA flange above the foam sponge"***).

*****Execution:**** The Pi 5 script immediately generates a unique `session_id` (e.g., `task_20260605_113000`), creates the corresponding folder, captures the raw frame from the D435i as `image.jpg`, and writes the transcribed text into `instruction.txt`.

*****Handshake:**** Send a network trigger packet to Module A containing the session ID: `{"status": "ready_for_pose", "session_id": "task_20260605_113000"}`.

2. Module A (Control & Execution) - The Kinematic Recorder

*****Action:**** The control script catches Module C's trigger and caches the `session_id` in memory.

*****Execution:**** The student manually guides the LARA arm via teach-pendant or drag-drop, aligning the ****Flange Tool Center Point (TCP)**** precisely above the target object.

* **Handshake:** Press Enter on the console to read LARA's current 6D Pose (X, Y, Z, Roll, Pitch, Yaw). Set the gripper parameter to a dummy value (e.g., 0), pack it into a JSON string, and send it back to Module C with the cached ID: {"session_id": "task_20260605_113000", "pose": [250, -100, 50, 90, 0, 0], "gripper": 0}.

3. Module B (Vision & AI) - The Verifier & Aggregator

* **Action:** Validate that the session_id inside pose_action.json matches the folder name perfectly.

* **CRITICAL RED LINE:** Verify image.jpg. It **MUST** be a completely clean image of the physical world. **DO NOT** draw any YOLO bounding boxes (OBB/SEG) on this image. The VLA backbone relies on its native attention mechanism; pre-rendered boxes will cause catastrophic model blindness during real-time inference.

* **LoRA Preparation:** Module B will freeze the entire 4B parameters of the Qwen3-VL backbone and only inject and train a lightweight bypass network (<1% of parameters). This enables full-precision training on 12GB RTX 4080 without Out-of-Memory (OOM) errors, converging within 1–2 hours.